Legal adapted cluster, score-trace cancellation, and terminal-Hessian frontier

TECT verification-first repository
claim `A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION`

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1. Purpose and scope

Five-round strategy

R-118 reduced the adapted A13 problem to a finite checklist but did not instantiate the legal production chart or decide whether its low Wiener chaoses cancel. This note carries out five coordinated proof rounds:

1. reconstruct the smallest genuinely adapted strict-triangular source chart and its legal output-cluster rule;
2. derive the exact zero/first-chaos score–trace criterion after frozen affine absorption;
3. test bare heat, canonical random-W, and scalar-model routes;
4. move from termwise owner basicness to the global terminal Hessian and the realized quotient norm;
5. synthesize the surviving route and record every failed or unresolved branch as a reusable decision.

What is proved. The smallest genuinely adapted strict-triangular test needs two independent source blocks and two visits; one root and two visits remain the smaller algebraic R-116 quotient fixture. The complete two-visit square-minus-trace packet telescopes exactly to its endpoints and owns the trace once.
For a finite-dimensional Gaussian endpoint $Y = b + A\xi + R$, with R orthogonal to zero and first chaos, the residual

$$\mathcal{R} = \operatorname{Re}\langle b + A\xi, R\rangle + \frac{1}{2}\|R\|^2 - \frac{1}{2}\delta\tau$$

has explicit zero and first Wiener projections. Their vanishing is equivalent to two exact aggregate trace identities. In chaos kernels these identities expose both the affine–quadratic contraction and every adjacent- chaos contraction. A bare Jacobian heat $\delta\tau = 2\operatorname{Re}\operatorname{Tr}(A^*DR) + \|DR\|_{\mathrm{HS}}^2$ leaves the strict mean debt $-\frac{1}{2}\sum_{n\geq 2}(n-1)n!\|r_n\|^2$ whenever $R \neq 0$.
The R-107 one-pair diagnostic nevertheless has an exact strictly positive canonical W and cost $2\sigma^8$. The R-115 scalar canonical coefficient is proved positive semidefinite on both covariance faces. No mixed-interior claim is made.
At fixed cutoff and positive floor, the Hessian of the complete renormalized terminal action factors automatically through endpoint synthesis. The quotient source norm makes this factorization unitary on the horizontal space, with no inverse singular-value or visit-count loss. Thus global vertical basicness is a chain-rule theorem; production reconstruction and spatial multiplier control, not another termwise primitive, are the remaining analytic burden.

What is not proved. The actual coefficient-level A1 complete cluster has not yet been shown to satisfy the two low-chaos identities. The diagnostic cluster $\{0, \pm 2k\}$ is not declared to be a complete production cluster. The mixed interior of the R-115 canonical W, the full trace/R-063 forest reconstruction, spatial H^2/L^6 quotient compatibility, multiplier variation, one-use source/sextic aggregation, $\mathrm{OVERLAP}_{\mathrm{src}}$, Nelson, removals, interacting measure, Sector-A closure, and tier promotion remain open.

The authorities used are R-063 for the coefficient forest, R-068 for the centered-form payment, R-082 for the production current, R-093 for directed source union, R-102 for the isolated-current curl, R-104 for fixed-chart assembly, R-107 for the coherent output cluster, R-108 for its signed normal form, R-115 for the stationary scalar theorem, and R-116–R-118 for endpoint quotients, trace horizons, and Gaussian range.

2. Round one: the legal adapted production chart

Let E_1, E_2 be independent finite Gaussian source blocks, with $\xi_j \sim N(0, \mathrm{I}_{E_j})$ and covariance-root maps S_j. Let h_1 be measurable with respect to the strict past before either new block, and let $h_2 = h_2(\xi_1)$ be measurable after the first reveal but before the second. The minimal genuinely adapted chart is

$$\begin{aligned} Z_0 &= S_1\xi_1 + S_2\xi_2, \\ Z_1 &= Z_0 + S_1 h_1, \\ Z_* &= Z_1 + S_2 h_2(\xi_1). \end{aligned} \tag{2.1}$$

One fresh root with two visits is enough for the algebraic R-116 endpoint quotient. It is not enough to test genuinely future feedback: a second independent reveal block is needed so that a later legal visit can depend on the first fresh root while remaining strict-past for its own root. This is a minimality statement about the filtration pattern, not a canonical claim about the real dimension of a covariance factorization.

For each endpoint $H \in \{0, 1, *\}$ form the full production current

$$\mathcal{J}_H = \left(\Xi_\Sigma(Z_H, \partial_i Z_H)\right)_{i=1}^{3}, \qquad x_H = \Pi_{\mathcal{C}} d_1 \mathcal{J}_H. \tag{2.2}$$

Here d_1 is the root-one martingale difference and $\Pi_{\mathcal{C}}$ is not an arbitrary singleton projection. Put a graph on retained output modes and connect two modes whenever they share a coordinate of the same fresh root, are joined by an R-063 contraction, or have a realized nonzero off-diagonal conditional covariance. A legal $\mathcal{C}$ is a union of connected components. In the scalar linear-row diagnostic this rule gives $\{0, +2k, -2k\}$. The rational $r/(\rho + e)$ row can generate all retained even harmonics, so the full production component may be much larger.

The current used in (2.2) is the R-082 full four-row current. For $\Psi = (u_1, u_2, \chi)$ and $\dot{\Psi} = (v_1, v_2, w)$, write

$$r = |u|^2, \quad \rho = r + |\chi|^2, \quad d = \rho + e,$$
$$\dot{r} = 2\,\mathrm{Re}(u^* v), \qquad \dot{\rho} = 2\,\mathrm{Re}(u^* v + \bar{\chi} w),$$
$$h = u_1 v_2 - u_2 v_1,$$

and

$$\Xi(\Psi, \dot{\Psi}) = \begin{pmatrix} \sqrt{c_0}\,\dot{r} \\ \sqrt{c_1}\left(\dot{r} - \frac{5}{9}\frac{r}{d}\dot{\rho}\right) \\ 2\sqrt{c_0 + c_1}\,\mathrm{Re}\,h \\ 2\sqrt{c_0 + c_1}\,\mathrm{Im}\,h \end{pmatrix}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \tag{2.3}$$

The heat lift is $\Xi_\Sigma(z, y)(r') = C(z + r')y$; averaging C before squaring is not the same operation.

Proposition 2.1 (complete two-visit telescope)

Let t_H denote the once-owned trace scalar at endpoint H. With $a = x_0$, $f = x_1 - x_0$, and $\iota = x_* - x_1$, the legal packet is

$$\begin{aligned} F_{2\mathrm{v}} = {} & \mathrm{Re}\langle a, f\rangle + \frac{1}{2}\|f\|^2 + \mathrm{Re}\langle a + f, \iota\rangle + \frac{1}{2}\|\iota\|^2 \\ & - \frac{1}{2}\{(t_1 - t_0) + (t_* - t_1)\} \\ = {} & \frac{1}{2}(\|x_*\|^2 - \|x_0\|^2) - \frac{1}{2}(t_* - t_0). \end{aligned} \tag{2.4}$$

Proof. Expand the two squares. The f–ι cross is generated exactly once by $\langle a + f, \iota\rangle$; all intermediate $\|x_1\|^2$ and t_1 terms cancel. This is an identity before expectation. $\square$

The raw t_H is the safe owner. Expanding its coefficient curvature by R-063 produces the P_3/P_1 and $P_4/P_2^{\mathrm{cross}}/\Sigma Q/P_0^{\mathrm{cross}}/\mathcal{R}_3$ forest. That forest is an alternative realization of the same trace increment and must not be added a second time.

3. Round two: exact score–trace cancellation

Condition on all strict-past data. Let ξ be a standard Gaussian vector in a finite-dimensional real Hilbert space E, and let H be a finite output Hilbert space. Decompose the complete endpoint output canonically as

$$Y = b + A\xi + R, \qquad \mathbb{E}R = 0, \qquad \mathbb{E}(R \otimes \xi) = 0. \tag{3.1}$$

The affine packet is handled by the frozen whole-output determinant. Put

$$\delta\tau = \tau - \|A\|_{\mathrm{HS}}^2, \qquad \mathcal{R} = \mathrm{Re}\langle b + A\xi, R\rangle + \frac{1}{2}\|R\|^2 - \frac{1}{2}\delta\tau. \tag{3.2}$$

Theorem 3.1 (aggregate zero and first chaos)

The zero and first Wiener projections of (3.2) are

$$\Pi_0\mathcal{R} = \frac{1}{2}(\mathbb{E}\|R\|^2 - \mathbb{E}\delta\tau), \tag{3.3}$$

$$\Pi_1\mathcal{R} = \langle m, \xi\rangle, \qquad m = \mathbb{E}\left[(DR)^*Y - \frac{1}{2}D\delta\tau\right]. \tag{3.4}$$

Consequently $\Pi_0\mathcal{R} = \Pi_1\mathcal{R} = 0$ if and only if

$$\boxed{\mathbb{E}\delta\tau = \mathbb{E}\|R\|^2}, \tag{3.5}$$

$$\boxed{\mathbb{E}D\delta\tau = 2\,\mathrm{Contr}(A, \mathbb{E}D^2R) + 2\mathbb{E}[(DR)^*R]}. \tag{3.6}$$

Proof. The two orthogonality conditions in (3.1) delete the expectation of both $\langle b, R\rangle$ and $\langle A\xi, R\rangle$, proving (3.3). Gaussian integration by parts gives the first-chaos coefficient as $\mathbb{E}D\mathcal{R}$. Differentiating (3.2) yields

$$D\mathcal{R} = A^*R + (DR)^*Y - \frac{1}{2}D\delta\tau. \tag{3.7}$$

The expectation of A^*R vanishes, which proves (3.4). Expanding $(DR)^*Y$ and integrating the affine term by parts gives (3.6). ☐

This calculation needs aggregated physical-space objects $Y, R, DR, \tau, D\tau$, not a separate Fourier expansion of every output mode. It does, however, need the actual production coefficients and all companions.

Corollary 3.2 (chaos-kernel form)

Write

$$R = \sum_{n\geq 2} I_n(r_n), \qquad \delta\tau = \sum_{n\geq 0} I_n(t_n), \tag{3.8}$$

with symmetric kernels and the convention $\mathbb{E}\|I_n(r_n)\|^2 = n!\|r_n\|^2$. Then (3.5)–(3.6) are equivalent to

$$t_0 = \sum_{n\geq 2} n!\|r_n\|^2, \tag{3.9}$$

$$(t_1)_i = 4\sum_j \langle Ae_j, r_2(i,j)\rangle + 2\sum_{n\geq 3} n!\langle r_{n-1}, r_n(\ cdot\ , i)\rangle. \tag{3.10}$$

The second sum is the adjacent-chaos debt. It is invisible if one inspects only the affine–quadratic contraction.

Proposition 3.3 (score representation)

Let $u = A^*R$ and define

$$S = \delta\tau - 2\,\mathrm{Re}\langle b, R\rangle - \|R\|^2 - 2\,\mathrm{Re}\,\mathrm{Tr}(A^*DR). \tag{3.11}$$

Then

$$\mathcal{R} = \delta(u) - \frac{1}{2}S. \tag{3.12}$$

Indeed $\delta(A^*R) = \langle A\xi, R\rangle - \mathrm{Tr}(A^*DR)$. Formula (3.12) shows why a signed score route remains natural, while (3.5)–(3.6) show that its low chaoses cannot be assumed away.

4. Round three: exact failed routes and positive diagnostics

Theorem 4.1 (bare Jacobian heat no-go)

Suppose R has no zero or first chaos and use only the bare Jacobian heat

$$\delta\tau_{\mathrm{bare}} = 2\,\mathrm{Re}\,\mathrm{Tr}(A^*DR) + \|DR\|_{\mathrm{HS}}^2. \tag{4.1}$$

Then the residual (3.2) has

$$\boxed{\Pi_0\mathcal{R}_{\mathrm{bare}} = -\frac{1}{2}\sum_{n\geq 2}(n-1)n!\|r_n\|^2}. \tag{4.2}$$

In particular it is strictly negative for every nonlinear $R \neq 0$.

Proof. Orthogonality kills the affine cross in expectation. The Gaussian derivative isometry gives

$$\mathbb{E}\|R\|^2 = \sum_{n\geq 2} n!\|r_n\|^2, \qquad \mathbb{E}\|DR\|_{\mathrm{HS}}^2 = \sum_{n\geq 2} n\,n!\|r_n\|^2,$$

and $\mathbb{E}\,\mathrm{Tr}(A^*DR) = 0$. Substitute in (3.3). ☐

Thus the output, low, and R-063 forest companions are not optional cosmetic terms. They must supply the exact missing debts if the complete A1 packet is to cancel.

Two one-dimensional fixtures separate the two low-chaos equations. With $A = 1$, $R = \epsilon H_2$, and $\delta\tau = 2\epsilon^2$, the mean vanishes but

$$\Pi_1\mathcal{R} = 2\epsilon H_1. \tag{4.3}$$

With $A = 0$, $R = \alpha H_2 + \beta H_3$, and $\delta\tau = 2\alpha^2 + 6\beta^2$, the mean again vanishes but

$$\Pi_1\mathcal{R} = 6\alpha\beta H_1. \tag{4.4}$$

These are method counterfixtures, not A1 counterexamples.

Proposition 4.2 (one-pair PSD coefficient)

For a standard two-dimensional Gaussian $x = (x_1, x_2)$, let $s = |x|^2$ and

$$P_\sigma = \frac{\sigma^4}{16}(s^2 - 4s). \tag{4.5}$$

With $H_2(u) = u^2 - 1$ and $H_4(u) = u^4 - 6u^2 + 3$, exact expansion gives

$$\begin{aligned} P_{sigma,2} &= \frac{\sigma^4}{4}\{H_2(x_1) + H_2(x_2)\}, \\ P_{\sigma,4} &= \frac{\sigma^4}{16}\{H_4(x_1) + H_4(x_2) + 2H_2(x_1)H_2(x_2)\}, \\ P_\sigma &= P_{\sigma,2} + P_{\sigma,4}. \end{aligned}$$

The R-118 canonical chaos inverse $W_\sigma = D^2 P_{\sigma,2} + D^2 P_{\sigma,4}/6$ therefore gives

$$W_\sigma(x) = \frac{\sigma^4}{24}\{2xx^T + (s+8)\mathrm{I}_2\} \tag{4.6}$$

satisfies

$$\delta^2 W_\sigma = 2P_\sigma, \qquad \mathbb{E}(\delta^2 W_\sigma)^2 = 2\sigma^8. \tag{4.7}$$

Its tangential and radial eigenvalues are respectively

$$\lambda_{\tan} = \frac{\sigma^4(s+8)}{24}, \qquad \lambda_{\mathrm{rad}} = \frac{\sigma^4(3s+8)}{24}. \tag{4.8}$$

Hence W_σ is strictly positive for $\sigma > 0$. The displayed chaos split proves canonicality, direct differentiation proves (4.7), and the fourth radial Gaussian moment gives the stated cost. This explains why the R-107 one-pair output cluster is a useful positive diagnostic even though universal PSD representation was refuted by R-118. It does not identify the complete rational production cluster.

Proposition 4.3 (stationary scalar boundary faces)

For the R-115 scalar model, the canonical coefficient is

$$W = D^2 F_2 + \frac{1}{3} D^2 F_3 + \frac{1}{6} D^2 F_4. \tag{4.9}$$

On the covariance face $c = 0$, the x block vanishes and the y block has

$$\lambda^{(0)}_{\tan} = \frac{48b + |y|^2 + 8}{96}, \qquad \lambda^{(0)}_{\mathrm{rad}} = \frac{48b + 3|y|^2 + 8}{96}. \tag{4.10}$$

On $c = 1$, the y block vanishes and the x block has

$$\lambda^{(1)}_{\tan} = \frac{12b + |x|^2 + 8}{24}, \qquad \lambda^{(1)}_{\mathrm{rad}} = \frac{12b + 3|x|^2 + 8}{24}. \tag{4.11}$$

Both nonzero blocks are strictly positive for $b \geq 0$. The formulas follow by differentiating F_2 and the radial fourth-chaos polynomial. The mixed interior $0 < c < 1$ includes the cubic cross block $D^2 F_3/3$; no exact global Gram or Schur proof has been obtained. Numerical reconnaissance is therefore not promoted to evidence.

5. Round four: global terminal Hessian and quotient geometry

Fix a finite cutoff, positive rational floor, and one R-104 source chart. Let $\mathcal{S}_\pi$ be its finite cylindrical source space and define

$$\begin{aligned} L_\pi h &= \sum_k S_k h_k, \\ Z_h &= S_\pi \xi + L_\pi h, \\ \mathcal{U}_{J,\pi}(h) &= \mathbb{E} V_J^{\mathrm{ren}}(Z_h). \end{aligned} \tag{5.1}$$

Theorem 5.1 (global terminal-Hessian factorization)

For cylindrical directions u, v,

$$D\mathcal{U}(h)[u] = \mathbb{E} D V_J^{\mathrm{ren}}(Z_h)[L_\pi u], \tag{5.2}$$

$$\boxed{D^2\mathcal{U}(h)[u,v] = \mathbb{E} D^2 V_J^{\mathrm{ren}}(Z_h)[L_\pi u, L_\pi v]}. \tag{5.3}$$

Therefore

$$\operatorname{Ker} L_\pi \subseteq \operatorname{Ker} D^2\mathcal{U}(h), \qquad D^2\mathcal{U}(h) = L_\pi^* B_{J,h} L_\pi \tag{5.4}$$

as a quadratic form on the realized endpoint range. Moreover

$$\mathcal{U}(h) - \mathcal{U}(0) = D\mathcal{U}(0)[h] + \int_0^1 (1-t) D^2\mathcal{U}(th)[h,h]\,\mathrm{d}t. \tag{5.5}$$

Proof. At fixed cutoff and positive floor, V_J^{ren} is a smooth finite-dimensional function with integrable polynomial/rational derivatives. Differentiate (5.1) under the Gaussian expectation. Since Z_h depends affinely on h, no second derivative of the synthesis map appears, proving (5.2)–(5.4). Apply the scalar integral Taylor formula to $t \mapsto \mathcal{U}(th)$ for (5.5). □

This theorem changes the route diagnosis. The complete terminal action is globally vertically basic without separately proving that each current, trace, or forest owner is basic. Termwise owners can still fail basicness; only their complete recombination is covered by (5.3).

Theorem 5.2 (realized quotient source norm)

For a bounded linear $L : \mathcal{S} \to E$, define on $\operatorname{Ran} L$

$$\|z\|_{q,L} = \inf_{Lh=z} \|h\|_{\mathcal{S}} \tag{5.6}$$

and complete this range to $\mathcal{Z}_L$. If $N = \operatorname{Ker} L$, then

$$\|Lh\|_{q,L} = \|P_{N^\perp} h\|_{\mathcal{S}}, \qquad U_L : N^\perp \to \mathcal{Z}_L, \qquad U_L h = Lh \tag{5.7}$$

is unitary. If a symmetric form K kills N, its quotient transport is

$$B = U_L (K|_{N^\perp}) U_L^{-1}, \qquad \|B\| = \|K|_{N^\perp}\|. \tag{5.8}$$

Absolute operator variation is preserved under the same unitary conjugation. Thus the algebraic quotient introduces no Moore–Penrose, small-singular-value, cutoff, or visit-count constant.

Proof. Every fibre has the unique minimum-norm representative in $N^{\perp}$, which proves (5.7). Formula (5.8) is unitary conjugation. □

The limitation is analytic: the quotient need not commute with spatial derivatives or sharp cubes. R-104 covariance-root maps smooth spatially. Using R-118's H^2/L^6 estimate still requires either commutation or a uniformly H^2/L^6-stable horizontal synthesis theorem.

6. Cartan checksum and owner completeness

The global one-form $D\mathcal{U}$ is exact, so its total curl vanishes. R-102's isolated rational-current fixture has

$$\partial_y \omega_x - \partial_x \omega_y = -\frac{40}{729}. \tag{6.1}$$

Consequently the complete trace, heat, low, and R-063 forest companions must contribute

$$\partial_y \omega_x^{\mathrm{comp}} - \partial_x \omega_y^{\mathrm{comp}} = +\frac{40}{729} \tag{6.2}$$

on the same fixture if the owner inventory is complete. Equation (6.2) is a required checksum derived from global exactness. It is not presented as an observed reconstruction and does not prove that a conditional subcluster or its canonical W is separately basic.

7. Round five: synthesized one-use frontier

The shortest surviving theorem target is now global. For $z = L_\pi h$, seek an owner-complete decomposition

$$D^2 V_J^{\mathrm{ren}}(X + tz)[z, z] = \langle Q_J, \Pi_{K_{J,t,h}} z \rangle + R^{+}_{J,t,h} + R^{\mathrm{sum}}_{J,t,h}, \tag{7.1}$$

such that

$$R^{+} \geq 0, \qquad \int_0^1 \mathbb{E}|R^{\mathrm{sum}}_{J,t,h}|\,\mathrm{d}t \leq C, \tag{7.2}$$

$$\sum_r |K_{J,t,h,r}| \leq K_0 \mathrm{I}, \qquad \sup_J \mathbb{E}\|Q_J\|^{10/3}_{H^{-11/10}} < \infty, \tag{7.3}$$

$$\|z\|^2_{H^2} \leq C_{\mathrm{CM}} \|h\|^2_{\mathcal{S}}, \qquad \|z\|_6^6 \leq C\{\|Z_h\|_6^6 + \|X\|_6^6\}. \tag{7.4}$$

If (7.1)–(7.4) hold with spatially constant fixed A1 generator matrices, R-118 gives one global Young step

$$|\langle Q, \Pi_K z \rangle| \leq \eta X + \zeta Y + C K_0^{10/3} \eta^{-11/6} \zeta^{-1/2} \|Q\|^{10/3}_{H^{-11/10}}. \tag{7.5}$$

This spends source energy and the terminal sextic once. If $K_r = K_r(x)$ is spatially varying, a pointwise spectral bound is insufficient; at least the multiplier variation

$$\sum_r |\langle u, \partial^\alpha K_r(x) v \rangle| \leq K_\alpha \|u\| \|v\|, \qquad |\alpha| \leq 2 \tag{7.6}$$

must be proved. The rough R-063 coefficient jets should therefore remain in Q_J, while the K_r should be assembled from fixed production generator matrices whenever possible.

8. Evidence map: routes, verdicts, and successor

Route	Verdict	Reusable conclusion / successor
Legal two-block chart	advanced	Two blocks and two visits are the minimum genuine adapted test; retain the contraction-connected whole output.
Low-chaos score–trace	advanced	Equations (3.5)–(3.6), or (3.9)–(3.10), are the exact production falsifiers.
Bare Jacobian heat	failed	Equation (4.2) leaves a strict nonlinear mean debt; complete low/output/forest companions are mandatory.
One-pair canonical W	advanced, diagnostic	It is strictly PSD with cost $2\sigma^8$, but $\{0, \pm 2k\}$ is not the rational production cluster.
R-115 canonical W	partial	Both covariance faces are PSD. Mixed interior remains unproved and is not promoted from numerical reconnaissance.
Termwise primitives	deprioritized	R-102 curl blocks the isolated current; global exactness requires companion curl $+40/729$.
Global terminal Hessian	advanced, preferred	Quotient factorization is automatic for the complete terminal action; reconstruct (7.1) and prove (7.3)–(7.6).
Raw spatial multiplier	boundary	Pointwise spectral control alone does not preserve the H^2 estimate.

This map prevents three repetitions: testing an illegal singleton output, assuming low-chaos cancellation from endpoint telescoping, and searching for an isolated current primitive after the nonzero-curl fixture.

9. Executable verification

The primary SymPy certificate derives the two-visit telescope, the general H_2/H_3 score–trace formulas, both counterfixtures, the bare-heat debt, the bivariate canonical coefficient from its chaos split and its cost, both scalar boundary faces, the directly differentiated terminal-Hessian quotient, and the inherited Cartan checksum. It passes 45/45 assertions.

A non-importing standard-library implementation reconstructs the same conclusions from exact rational univariate and multivariate polynomial routines, including independently reconstructed chaos inverses, face Hessians, and the Hessian of the composed endpoint potential. It passes 28/28 assertions. The integrated verifier reruns both children, compares deterministic artefacts, checks every authority/source/note/PDF hash, validates the route and no-overclaim surfaces, and fails closed on stale output.

10. Devil's-advocate review

1. **Objection: the chart (2.1) proves the full A1 cluster is only $\{0, \pm 2k\}$. UPHELD as false.** That set belongs to the scalar linear-row diagnostic. Rational output harmonics and realized covariance connections may force the whole retained output component.

2. **Objection: endpoint telescoping proves low-chaos cancellation. UPHELD as false.** Equations (4.3)–(4.4) exhibit independent first- chaos debts after exact mean centering. The production trace must satisfy both (3.5) and (3.6).

3. **Objection: the Jacobian heat is the trace, so it must cancel. UPHELD as false.** Equation (4.2) is strictly negative for every nonlinear residual. Additional low/output/R-063 companions are load-bearing.
4. **Objection: R-118's signed universal counterfixture rules out a positive production coefficient. DISMISSED.** Proposition 4.2 is an exact positive diagnostic. Positivity is packet-specific, not universal.
5. **Objection: positive covariance faces prove the mixed R-115 interior by continuity. UPHELD as false.** Boundary positivity has no uniform gap in inactive directions, and the cubic cross block appears in the interior. A Gram/Schur proof is still required.
6. **Objection: Theorem 5.1 proves every local owner is vertically basic. UPHELD as false.** It proves only the fully recombined global action. Owner exchange and the $\pm 40/729$ curl checksum remain essential.
7. **Objection: the quotient norm automatically gives the physical H^2/L^6 bounds. UPHELD as false.** The unitary quotient is in source geometry. Spatial derivative and sharp-cube compatibility are a separate synthesis theorem.
8. **Objection: pointwise $\sum_r |K_r(x)| \leq K_0 \mathrm{I}$ is enough. UPHELD as false.** Differentiated H^2 estimates see derivatives of $K_r(x)$; (7.6) or an equivalent multiplier theorem is necessary.
9. **Objection: the executable constants may be pasted or hide a sign, factor, convention, or scaling error. VALID-with-mitigation.** The primary child derives the one-pair coefficient from its H_2/H_4 split and differentiates $V(Lh)$ directly. The non-importing child reconstructs the chaos inverse, both face Hessians, and the composed Hessian with exact rational polynomials. The σ^4/σ^8 homogeneities and $\sigma \downarrow 0$, $b = 0$ limits agree. This is fixed-cutoff evidence; no cutoff convergence is inferred.
10. **Objection: R-119 closes either A13 gate or Sector A. DISMISSED.** It narrows the gates and removes one global algebraic uncertainty. The production identities and uniform analytic estimate remain open, so the tier stays T4.

11. Result footer

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Result ID: A13-CLASSII-LEGAL-ADAPTED-CLUSTER-SCORE-TRACE-
  TERMINAL-HESSIAN-FRONTIER (R-119).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: Prop. 2.1; Thm. 3.1, Cor. 3.2, Prop. 3.3; Thm. 4.1,
  Props. 4.2-4.3; Thms. 5.1-5.2; checksum (6.1)-(6.2);
  conditional one-use target (7.1)-(7.6).
Scope: fixed finite cutoff/floor and cylindrical Gaussian source charts;
  complete-terminal-action quotient; diagnostics are not full A1.
Dependencies: R-063,068,082,093,102,104,107,108,115,116,118.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_legal_adapted_cluster_score_trace_
  terminal_hessian_frontier_verify.py
Expected output: R-119 PASS 90/90; children 45/45 and 28/28; all pins valid.
Falsification gate: any failed telescope, projection, chaos coefficient,
  bare-heat/W/face/quotient/checksum/hash assertion, or use beyond scope.
Tier before / after: T4 / T4.
No-overclaim statement: full A1, mixed PSD, forest, spatial synthesis,
  one-use, OVERLAP_src, Nelson, removals, measure, Sector-A/T5-T7 remain open.
Next required action: prove (7.1)-(7.6) for fixed A1 generators; use R-118 once.
```